

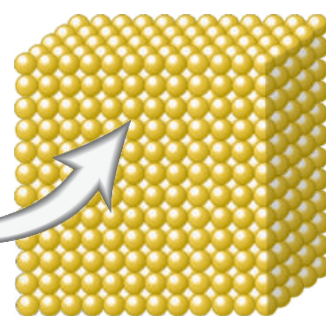


UNIT 01

FUNDAMENTALS OF CHEMISTRY



One-ounce gold coin



100,000,000,000,000,000,000,000
(10^{23}) gold atoms

SHORT QUESTIONS

1) Define chemistry?

“Chemistry is that branch of science which deals with the structure, properties, composition of matter and of the changes that occurs in matter.”

2) Write the contributions of Greek scientists in the field of chemistry?

S. #	NAME	TIME PERIOD	CONTRIBUTION
1.	Aristotle	384-322 B.C	He proposed idea of a substance as a combination of matter. He gave the theory of the Four Elements, fire, water, earth, air.
2.	Plato	347-428 B.C	He proposed term 'elements' as a composition of matter with a specific shape.
3.	Democritus	357-460 B.C	He proposed the idea of an atom as an indivisible particle of matter.

3) What are the contributions of Muslim scientists in the development of chemistry?

S. #	NAME	TIME PERIOD	CONTRIBUTION
1.	Jabir Ibne hayan جابر ابن حَيَّان	721-803 A.D	I. Invented experimental method of nitric acid (HNO_3), hydrochloric acid (HCl) and white lead (PbCO_3), $\text{Pb}(\text{OH})_2$. II. Extraction of metals for their ores and dyeing of clothes.
2.	Al razi الرازي	862-930 A.D	He prepared ethyl alcohol ($\text{C}_2\text{H}_5\text{OH}$) by fermentation process.
3.	Al-Beruni البيروني	973-1048 A.D	He determined the densities of different substance.
4.	Ibne-sina ابن سينا	980-1037 A.D	He contributed in medicines, philosophy, and astronomy.

4) What is the importance of chemistry? OR Write down about the applications of chemistry in daily life. [2025]

IMPORTANCE OF CHEMISTRY IN OUR DAILY LIFE

Chemistry is all around us in everything we see and use. Here's how:

- The air you breathe is a mixture of gases like oxygen and nitrogen.
- The book you're reading is made from wood, which has a chemical substance called **cellulose**.
- Your clothes may be made from synthetic materials like **nylon or terylene**, which are chemicals.
- The seat you are sitting on is probably made of plastic, a type of chemical material.
- The room you are in has things like cement, plastic, concrete, and glass, all of which are chemicals.

This shows us that chemistry is everywhere, and it affects all parts of our daily life. That's why it's important to study chemistry!



▲ Chemistry is central to our understanding of the world around us.

5) Define the term atom and all those terminologies which is concerned with the atom?

ATOM

It is a smallest unit of an element and enter in a chemical reaction.

OR

An atom is the smallest particle of an element that takes part in chemical change.

ATOMIC NUMBER OR PROTON NUMBER (Z)

The number of protons in the nucleus of an atom of an element is called as Atomic number. It is equal to the number of electrons orbiting round the nucleus of that neutral atom. It is denoted by "Z".

MASS NUMBER OR NUCLEON NUMBER (A)

The sum of proton and neutron present in the nucleus of an atom is known as mass number.

$$\therefore A = P + n \text{ OR } A = Z + n$$

ATOMIC MASS

The atomic mass of a single atom is simply its total mass and is typically expressed in atomic mass units or amu.

RELATIVE ATOMIC MASS (ATOMIC WEIGHT)

The relative atomic mass (A_r) of an element is defined as the ratio of the 'average mass of an atom' to 1/12th the mass of a $^{12}_6\text{C}$ atom."

6) What is substance and element?

SUBSTANCE

The purest form of matter is known as substance. They contain only one type of atom or molecule.

ELEMENT

An element is a substance which cannot be split into two or more simpler substances by ordinary chemical means.

OR

It is a pure substance or least complex form of matter and by simple chemical reaction not converted in other simpler substance.

7) What is matter? How many states of matter are there?

MATTER

It is anything, which has weight and occupies space or simply all the things which are present in the universe is matter. It can exist in five states, solids, liquid, gas, plasma and BEC.

8) What is valency?

VALENCY

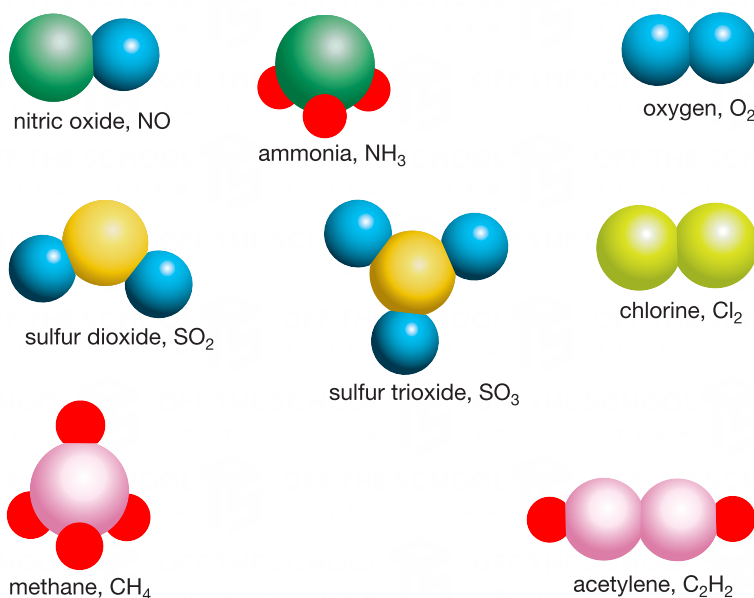
The combining power of an element with the other elements is called valency.

- Some elements contain commonly single valency [lithium +1].
- Some possess dual valency means they contain both +1 and -1 [hydrogen].
- Some elements show variable valencies like transition elements [copper +1, +2 and +3].

9) What do you know about the compound and molecule?

MOLECULE

A molecule is the smallest particle of compound and is made up of groups of atoms. Molecules can be monoatomic, diatomic and poly atomic.^[1]



▲ The image shows the atoms present in molecules of various compounds.

① Examples are NO, NH₃, O₂ respectively. See more examples on your text book page 09.

COMPOUND

A compound is a pure substance made of molecules with atoms from different elements, chemically combined.

To show the atoms present in a molecule of a compound we use its chemical formula.

10) Discuss about the mixture and their types?

MIXTURE

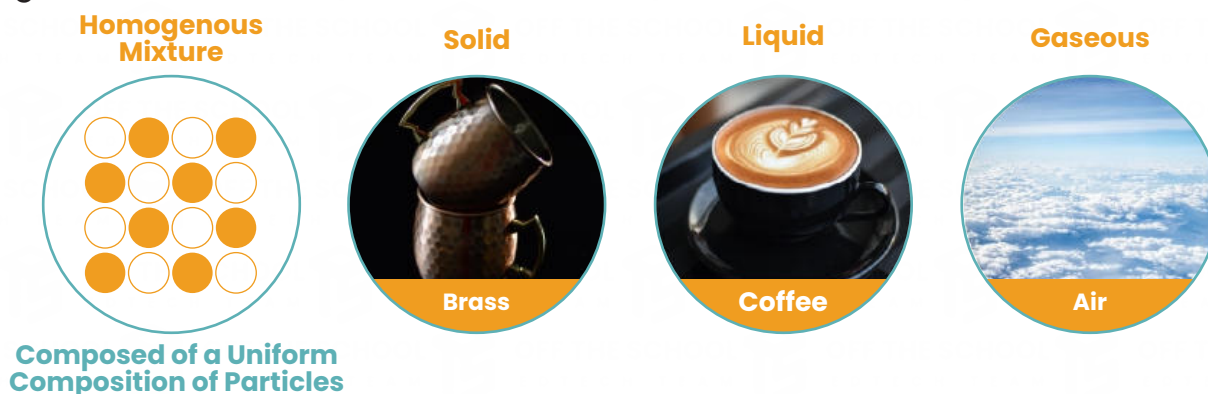
When two or more substances are physically combined together is known as mixture. Mixture can be separated in their parent elements by the simple physical methods like distillation, evaporation, crystallization and magnetization etc.

There are two types of mixture usually;



1. HOMOGENEOUS (huh-mo-juh-nuhs) MIXTURE

One that is uniform (same) in composition and properties throughout is said to be a homogeneous mixture or a solution^[1].



2. HETEROGENEOUS MIXTURE

One that is non-uniform in composition and properties throughout. e.g, soil, rock, pizza, chocolate chip cookies etc.



11) Define empirical and molecular formula?

EMPIRICAL FORMULA

The formula which show the least number of atoms present in a molecule.

MOLECULAR FORMULA

The formula which show the actual number of atoms present in a molecule.

NAME OF COMPOUND	MOLECULAR FORMULA	EMPIRICAL FORMULA
Hydrogen peroxide	H_2O_2	HO
Benzene	C_6H_6	CH

12) How can we define the term molecular and formula mass?

MOLECULAR MASS

"The sum of all atomic masses presents in a molecule."

FORMULA MASS

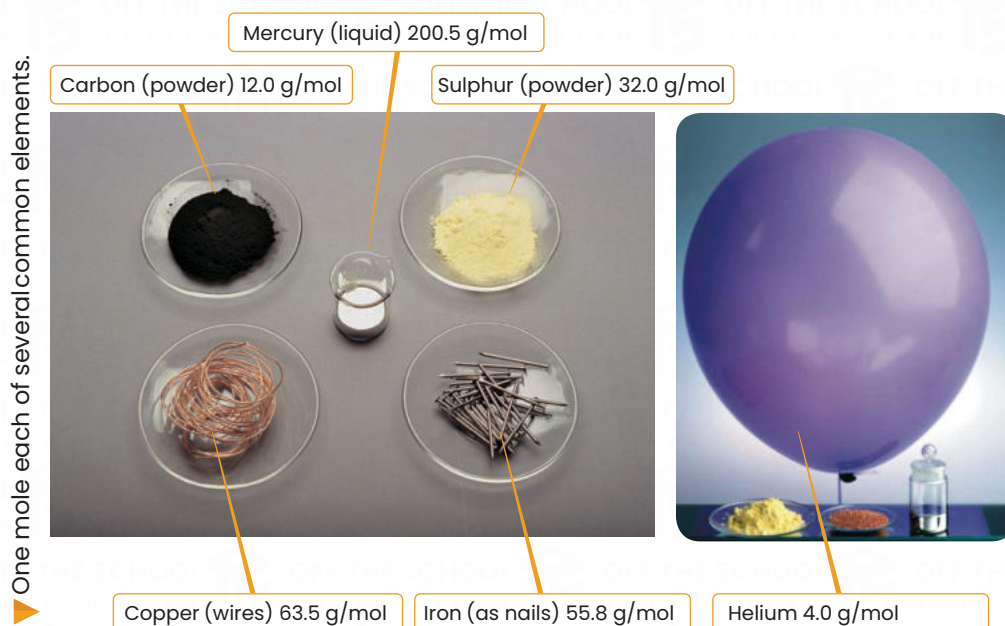
"The sum of all atomic masses presents in a formula unit of ion."

13) Define mole and the terminologies related to the mole?

MOLE

Gram atomic mass or gram molecular and gram formula mass of any substance is called Mole. **OR**

A mole is defined as the amount of substance which contain the **Avogadro number** of particles.



AVOGADRO'S NUMBER

The Avogadro number (or Avogadro constant) is defined as the number of atoms in 12 grams of the C-12 isotope. **OR**

One mole of a substance contains 6.02×10^{23} number of particles this number is called Avogadro number. ^[1]

1 Thinking About Avogadro's Number

Avogadro's number (6.02×10^{23}) is an enormously large number and practically inconceivable in terms of ordinary experience. Suppose we were counting garden peas instead of atoms. If the typical pea had a volume of about 0.1 cm^3 the required pile of peas would cover the **Islamic republic of Pakistan** to a depth of about 6 km (4 miles).

What is a mole in chemistry?



What is a mole?

One mole is the amount of substance that contains exactly $6.02214076 \times 10^{23}$ atoms, molecules or ions. This number is also known as 'Avogadro's number'. It's named after Italian scientist Amedeo Avogadro (left), a suggestion put forward by French scientist Jean Perrin to recognise Avogadro's work. 'Mole' derives from molecule – it's not related to the animal.



602,214,076,000,000,000,000,000

The number of atoms, molecules or ions in one mole of a substance

Atoms, molecules and ions are very small and impossible for chemists to count. Using the mole makes it easier to talk about amounts of substances involved in reactions, by relating the mass of a substance to its atomic or molecular mass.

Amount of substance = mass (g) $\div$ mass of 1 mole (g mol^{-1})



Water



Oxygen



Table salt



Gold



Helium

One mole contains a different mass for different substances

This makes sense if you think about it. Different substances will have atoms, molecules or ions which have different masses. Gold atoms have a greater mass than iron atoms, so the mass contained in one mole of gold atoms is greater.

Coins and moles: A useful analogy

Using moles to express amount of substance is analogous to weighing coin rolls to estimate the number of coins. In this analogy, the value of the coins is like mass (different for different coins), the number of coins is like number of atoms, and the rolls of coins are like moles of atoms.

MOLAR MASS

The total mass in grams present in one mole of any substance is called molar mass.



(a) carbon (graphite)

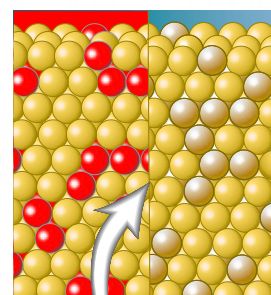
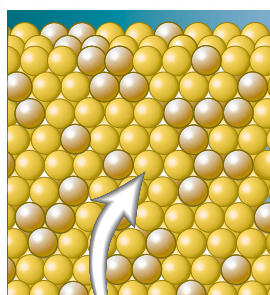
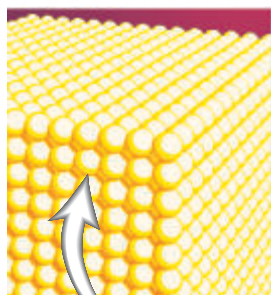
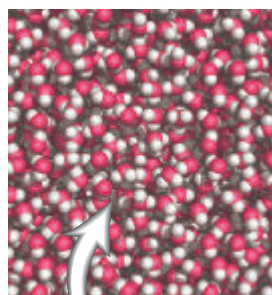
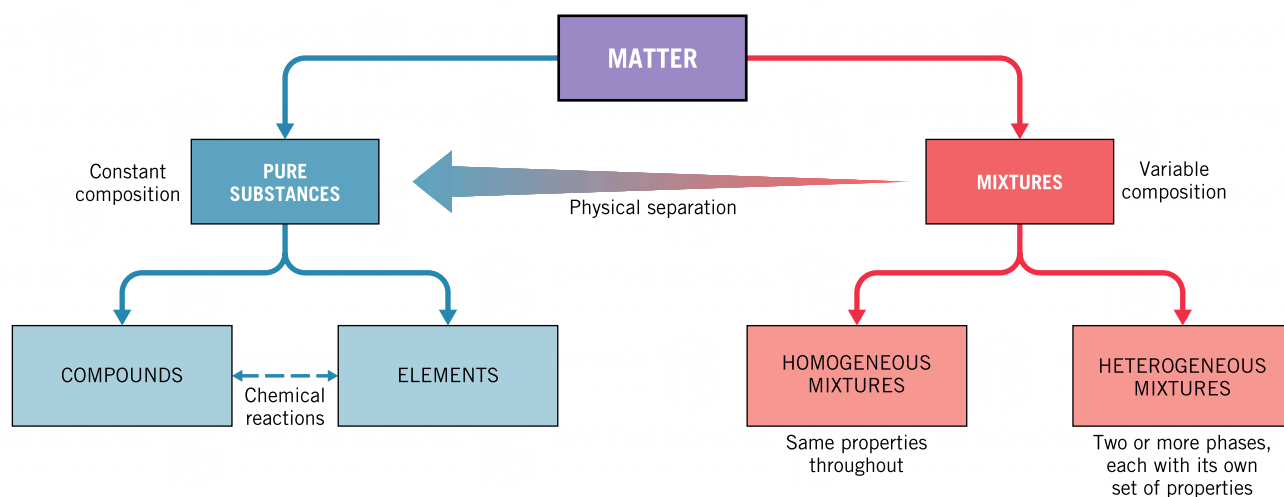


(b) iron (nails)



(c) copper (wire)

▲ Molar Mass Shown is approx. one molar mass of each of three elements.



▲ A scheme for classifying matter according to its composition. From left to right: The coin is pure gold, an element. Water is a compound. The pendant is made of rose gold, a homogeneous mixture of gold and copper; all parts of the pendant have the same composition. The ring at the far right is a heterogeneous mixture of rose gold and white gold (which contains gold and silver). Rose gold and white gold have different compositions.

LONG QUESTIONS

1) Write the branches of chemistry? OR Define chemistry and list any five branches of it and explain them. (2022)

1. PHYSICAL CHEMISTRY

Physical Chemistry is the branch of chemistry that looks at how the makeup of matter affects its properties and how those properties change during reactions or processes.

2. ORGANIC CHEMISTRY

Organic Chemistry is the study of carbon and hydrogen (hydrocarbons) and their derivatives with some exemption.

3. INORGANIC CHEMISTRY

Inorganic chemistry deals with the study of all elements and their compounds except hydrocarbons and their derivatives.

4. BIOCHEMISTRY

It is the branch of chemistry in which we study the structure, composition, and chemical reactions of substances found in living organisms.

5. INDUSTRIAL OR APPLIED CHEMISTRY

The branch of chemistry that deals with the manufacturing of chemical compounds on commercial scale, is called industrial chemistry.

6. NUCLEAR CHEMISTRY

Nuclear Chemistry is the branch of chemistry that deals with the radioactivity, nuclear processes and properties. The main concern of this branch is with the atomic energy.

7. ENVIRONMENTAL CHEMISTRY

It is the branch of chemistry in which we study about components of the environment and the effects of human activities on the environment.

8. ANALYTICAL CHEMISTRY

It is the branch of science that deal with the qualitative and quantitative analysis of matter.

9. MEDICINAL CHEMISTRY

This branch of chemistry studies drugs and the creation of chemical compounds.

10. QUANTUM CHEMISTRY

This branch of chemistry focuses on the study and experiments of physical models in chemical systems, also known as molecular quantum mechanics.

11. GREEN CHEMISTRY

The branch of chemistry which deals with study of processes and designing products, which are composed of less hazardous (less harmful) substances. It is also known as sustainable chemistry.



Previous Year's Questions

The branch of chemistry in which purple iodine vapours appear when solid iodine is heated, is called:

- Environmental Chemistry
- Biochemistry
- Medicinal chemistry
- Inorganic chemistry

[2025]



Previous Year's Questions

The following branch belongs to which branch of chemistry? Identify:

- The chlorofluoro compounds are responsible for depletion of ozone layer.
- That investigates the properties of ammonia gas.
- That find out the quantity and quality of Paracetamol in Panadol tablet.

[2024]

Some elements with symbols derived from their Latin names

USUAL NAME	LATIN NAME	SYMBOL
Copper	Cuprum	Cu
Gold	Aurum	Au
Iron	Ferrum	Fe
Lead	Plumbum	Pb
Mercury	Hydragyrum	Hg
Potassium	Kalium	K
Silver	Argentum	Ag
Sodium	Natrium	Na
Tin	Stannum	Sn
Antimony	Stibium	Sb

LINK TO LEARNING

Many compounds break down when heated. This site (<http://openstaxcollege.org/l/16mercury>) shows the breakdown of mercury oxide, HgO. You can also view an example of the photochemical decomposition of silver chloride (<http://openstaxcollege.org/l/16silvchloride>) (AgCl), the basis of early photography.

LINK 01



LINK 02

**2) What are molecules? Discuss its types with examples.(2024)****MOLECULE**

A molecule is the smallest particle of compound and is made up of groups of atoms.

TYPES OF MOLECULE ON THE BASIS OF NO.OF ATOMS

Organic Chemistry is the study of carbon and hydrogen (hydrocarbons) and their derivatives with some exemption.

1. Monoatomic Molecules:

These molecules consist of a single atom. Examples include noble gases like helium (He), neon (Ne), and argon (Ar), which exist independently in nature.

2. Diatomic Molecules:

These molecules are made up of two atoms, either of the same or different elements. Common examples include oxygen (O₂), hydrogen (H₂), and nitrogen (N₂).

3. Triatomic Molecules:

As the name suggests, triatomic molecules contain three atoms. Water (H_2O) and carbon dioxide (CO_2) are prime examples.

4. Polyatomic Molecules:

Polyatomic molecules consist of many atoms. Examples include methane (CH_4), sulfuric acid (H_2SO_4), and glucose ($\text{C}_6\text{H}_{12}\text{O}_6$).

TYPES OF MOLECULE ON THE BASIS OF NATURE OF ATOMS

5. Homoatomic Molecules:

These molecules are composed of atoms of the same element. Oxygen (O_2), ozone (O_3), and sulfur (S_8) are common examples.

6. Heteroatomic Molecules:

Heteroatomic molecules are made up of atoms of different elements. Water (H_2O), carbon dioxide (CO_2), and ammonia (NH_3) are all examples of heteroatomic molecules.

3) Differentiate between Molecular or Empirical formula.

(2025)

EMPIRICAL FORMULA	MOLECULAR FORMULA
DEFINITION	
1. It shows the simplest ratio of atoms of each element in a compound.	It shows the actual number of atoms of each element in a molecule of the compound.
REPRESENTATION	
2. Represents the compound in the simplest whole number ratio.	Represents the actual number of atoms in a molecule.
EXAMPLE	
3. For hydrogen peroxide (H_2O_2), the empirical formula is HO because the ratio of hydrogen to oxygen is 1:1.	For hydrogen peroxide, the molecular formula is H_2O_2 , which shows the actual number of hydrogen and oxygen atoms.
USEFULNESS	
4. Useful for showing the simplest form of the compound, especially in large compounds.	Useful for understanding the actual structure and size of the molecule.

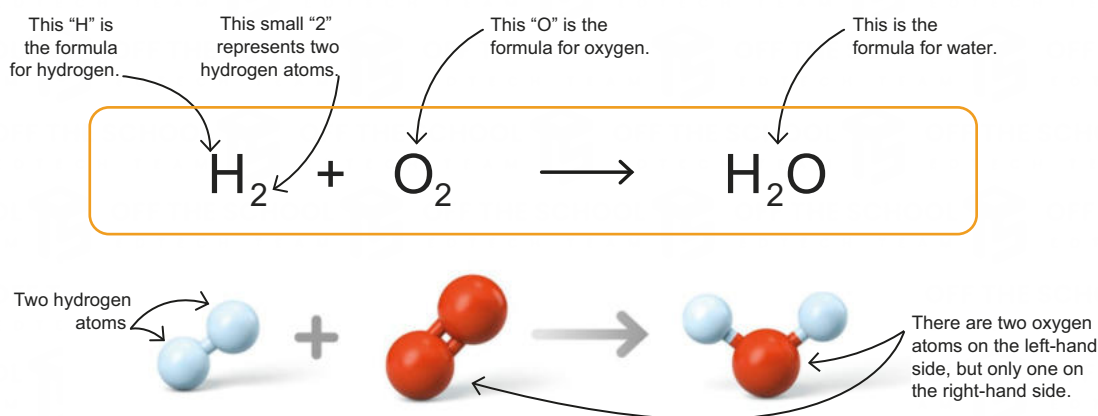
BALANCING EQUATIONS

An unbalanced equation

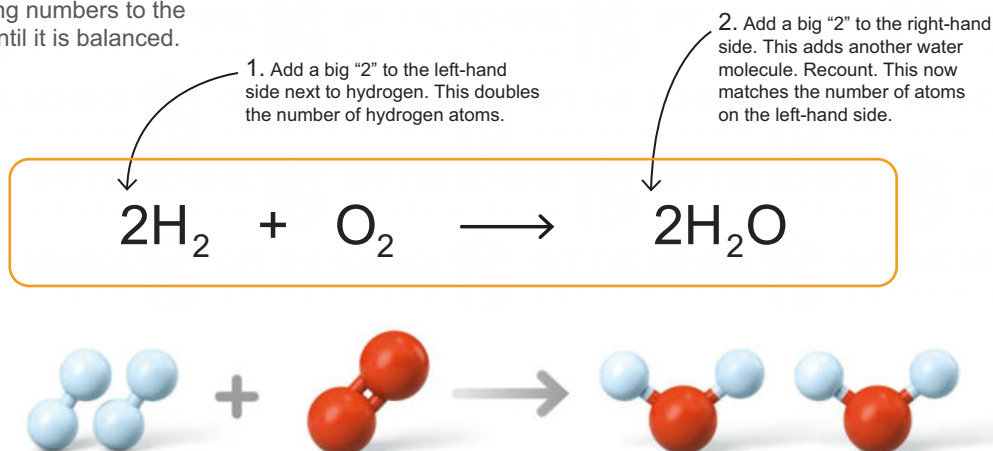
This equation shows the elements hydrogen and oxygen reacting to make water. It is an unbalanced equation, because it has two oxygen atoms on the left but only one on the right.

Key Facts

- ✓ Equations must be balanced, with the same number of atoms on both sides, because atoms are never lost or gained during reactions.
- ✓ Unbalanced equations can be balanced by adding numbers in front of the formulas, until both sides have the same number of atoms.



How to balance an equation
Keep adding numbers to the equation until it is balanced.



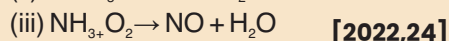
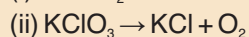
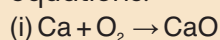
A balanced equation ▲

It is a balanced equation, because it has two oxygen atoms on the left and two on the right. And four hydrogen atom on both sides.

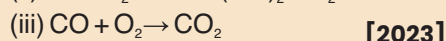
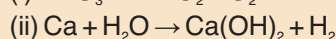
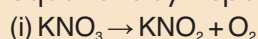
Previous Year's Questions



Balance the following chemical equations:



Balance the following chemical equations by inspection method:



NUMERICAL OF MASS NUMBER OR NUCLEAR EQUATION

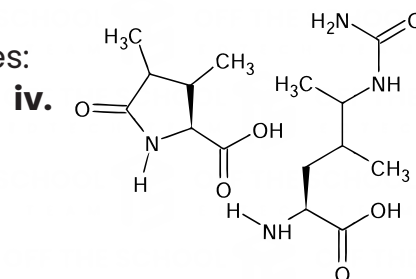
1. Calculate the mass number of an atom that contains 15 protons and 16 neutrons.
2. An element has an atomic number of 11 and a mass number of 2, How many neutrons does this atom have?
3. What is the mass number of an atom that contains 26 protons, 26 electrons, and 30 neutrons?
4. An atom has an atomic number of 19 and a mass number of 39. How many protons does this atom have?
5. How many electrons does an atom with an atomic number of 8 and a mass number of 16 have?
6. Determine the mass number of an atom that has 17 protons, 17 electrons, and 18 neutrons.
7. An atom has an atomic number of 6 and a mass number of 12. How many electrons does this atom have?
8. Complete the following vacant spaces in the following table:

ATOMIC NUMBER	MASS NUMBER	NUMBER OF ELECTRONS	NUMBER OF PROTONS	NUMBER OF NEUTRONS
1	1	1	1	0
2	4	2	2	
7			7	
12				
5		5		6
19				
8				8
40				
11			11	

NUMERICAL OF MOLE

CASE # 01: MOLE-MASS CALCULATION:

- 1) Calculate the no. of moles in 30 g of Na?
- 2) How many moles of calcium chloride should be produced by 50 g?
- 3) If the molecular mass of a gas is 92.009 g/mol, then what should be its mass if it contains 2.33×10^{-4} mole?
- 4) Calculate the mass of the following when it contains 1.0012 moles:
 i- CaCO_3 ii- BaCl_2
- 5) Calculate the molecular mass of the following molecules:
 i. $\text{C}_6\text{H}_{12}\text{O}_6$ ii. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ iii. CH_3COOH
- 6) Find out the mass of 3 moles of oxygen atoms.
- 7) Calculate the molecular masses of the following:



1.	Potassium permanganate KMnO_4	5.	Aluminum sulfate $\text{Al}_2(\text{SO}_4)_3$	9.	Magnesium nitrate $\text{Mg}(\text{NO}_3)_2$
2.	Acetic acid CH_3COOH	6.	Calcium hydroxide $\text{Ca}(\text{OH})_2$	10.	Potassium dichromate $\text{K}_2\text{Cr}_2\text{O}_7$
3.	Sulfur hexafluoride SF_6	7.	Ammonium phosphate $[(\text{NH}_4)_3\text{PO}_4]$	11.	Glucose-6-phosphate $\text{C}_6\text{H}_{13}\text{O}_9\text{P}$
4.	Aspirin $\text{C}_9\text{H}_8\text{O}_4$	8.	Iron (III) oxide Fe_2O_3	12.	Sulfuric acid H_2SO_4

CASE #02: AVOGADRO'S NUMBER:

1. Calculate the no. of atoms present in 9.66 g of calcium?
2. If a mixture of a gas contains 2.5 moles and having a mass of 8.00 grams. What will be its molecular mass and also find the no. of atoms present in the mixture of a gas?
3. Calculate the number of ions present in 0.003 g of Na^+Cl^- ?
4. Calculate the no. of molecules in 10 gm of H_2SO_4 ?
5. How many molecules of water are present in half mole of water?
6. Calculate the no. of molecules and moles present in the following?
 - a) 16g of H_2CO_3
 - b) 20g of $\text{C}_6\text{H}_{12}\text{O}_6$

ATOMIC MASSES OF THE ELEMENTS:

K= 39 amu	Mn= 54.94 amu	O= 16.00 amu	Al= 27 amu
S= 32.0 amu	Mg= 24.31 amu	N= 14.01 amu	C= 12.0 amu
H= 1.0 amu	Og = 294 amu	Cr= 52.00 amu	Ba= 137.33
F= 19.00 amu	Na= 23 amu	P= 30.97 amu	Fe= 55.85 amu
He= 04 amu	Cu= 63.5 amu	Ca= 40 amu	Cl= 35.5 amu

WORKSHEET BALANCE THE CHEMICAL EQUATIONS

LEVEL 01:

- 1) $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$
- 2) $\text{KClO}_3 \rightarrow \text{KCl} + \text{O}_2$
- 3) $\text{NaCl} + \text{F}_2 \rightarrow \text{NaF} + \text{Cl}_2$
- 4) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
- 5) $\text{C} + \text{S}_8 \rightarrow \text{CS}_2$
- 6) $\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}$
- 7) $\text{N}_2 + \text{O}_2 \rightarrow \text{N}_2\text{O}_5$
- 8) $\text{Al} + \text{S}_8 \rightarrow \text{Al}_2\text{S}_3$
- 9) $\text{Cs} + \text{N}_2 \rightarrow \text{Cs}_3\text{N}$
- 10) $\text{Mg} + \text{Cl}_2 \rightarrow \text{MgCl}_2$
- 11) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- 12) $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- 13) $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- 14) $\text{P} + \text{O}_2 \rightarrow \text{P}_2\text{O}_5$
- 15) $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$
- 16) $\text{Ag}_2\text{O} \rightarrow \text{Ag} + \text{O}_2$
- 17) $\text{S}_8 + \text{O}_2 \rightarrow \text{SO}_3$
- 18) $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{O}_2$
- 19) $\text{K} + \text{MgBr}_2 \rightarrow \text{KBr} + \text{Mg}$

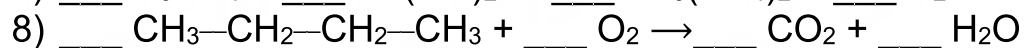
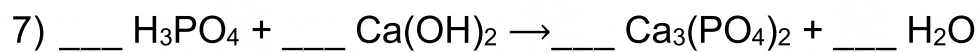
- 20) $\text{H}_2\text{O} + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$
 21) $\text{NaBr} + \text{CaF}_2 \rightarrow \text{NaF} + \text{CaBr}_2$
 22) $\text{H}_2\text{SO}_4 + \text{NaNO}_2 \rightarrow \text{HNO}_2 + \text{Na}_2\text{SO}_4$

LEVEL 02:

- 1) $\text{H}_3\text{PO}_4 + \text{KOH} \rightarrow \text{K}_3\text{PO}_4 + \text{H}_2\text{O}$
 2) $\text{K} + \text{B}_2\text{O}_3 \rightarrow \text{K}_2\text{O} + \text{B}$
 3) $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
 4) $\text{Na} + \text{NaNO}_3 \rightarrow \text{Na}_2\text{O} + \text{N}_2$
 5) $\text{NaOH} + \text{H}_2\text{CO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$
 6) $\text{KOH} + \text{HBr} \rightarrow \text{KBr} + \text{H}_2\text{O}$
 7) $\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}$
 8) $\text{Al}(\text{OH})_3 + \text{H}_2\text{CO}_3 \rightarrow \text{Al}_2(\text{CO}_3)_3 + \text{H}_2\text{O}$
 9) $\text{Rb} + \text{RbNO}_3 \rightarrow \text{Rb}_2\text{O} + \text{N}_2$
 10) $\text{C}_6\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 11) $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$
 12) $\text{C}_{10}\text{H}_{22} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 13) $\text{Al}(\text{OH})_3 + \text{HBr} \rightarrow \text{AlBr}_3 + \text{H}_2\text{O}$
 14) $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 15) $\text{Li} + \text{AlCl}_3 \rightarrow \text{LiCl} + \text{Al}$
 16) $\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 17) $\text{NH}_4\text{OH} + \text{H}_3\text{PO}_4 \rightarrow (\text{NH}_4)_3\text{PO}_4 + \text{H}_2\text{O}$
 18) $\text{Rb} + \text{P} \rightarrow \text{Rb}_3\text{P}$
 19) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 20) $\text{Al}(\text{OH})_3 + \text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
 21) $\text{Na} + \text{Cl} \rightarrow \text{NaCl}$
 22) $\text{Rb} + \text{S}_8 \rightarrow \text{Rb}_2\text{S}$
 23) $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$
 24) $\text{Li} + \text{H}_2\text{O} \rightarrow \text{LiOH} + \text{H}_2$
 25) $\text{NH}_3 + \text{O}_2 \rightarrow \text{N}_2 + \text{H}_2\text{O}$
 26) $\text{FeS}_2 + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$
 27) $\text{C} + \text{SO}_2 \rightarrow \text{CS}_2 + \text{CO}$
 28) $\text{Pb}(\text{OH})_2 + \text{HCl} \rightarrow \text{H}_2\text{O} + \text{PbCl}_2$
 29) $\text{FeCl}_3 + \text{NaOH} \rightarrow \text{Fe}(\text{OH})_3 + \text{NaCl}$

LEVEL 03: (Problems which are the most difficult. If you can balance these, you can balance any equation given in a class).

- 1) $\text{S} + \text{HNO}_3 \rightarrow \text{H}_2\text{SO}_4 + \text{NO}_2 + \text{H}_2\text{O}$
 2) $\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO} + \text{H}_2\text{O}$
 3) $\text{Ca}_3(\text{PO}_4)_2 + \text{SiO}_2 + \text{C} \rightarrow \text{CaSiO}_3 + \text{CO} + \text{P}$
 4) $\text{AlBr}_3 + \text{K}_2\text{SO}_4 \rightarrow \text{KBr} + \text{Al}_2(\text{SO}_4)_3$
 5) $\text{HCl} + \text{CaCO}_3 \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
 6) $\text{HNO}_3 + \text{NaHCO}_3 \rightarrow \text{NaNO}_3 + \text{H}_2\text{O} + \text{CO}_2$

**LINK TO LEARNING**

Use this interactive tutorial (<http://openstaxcollege.org/l/16BalanceEq>) for additional practice balancing equations.

LINK 03

**Science may set limits to
knowledge, but should not set
limits to imaginations.**

